

NICHOLSON ENGINEERING ASSOCIATES, INC.

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PROJECT ADDRESS:	14704 CARNATION AVENUE, TAMPA, FL 33613
COUNTY:	HILLSBOROUGH COUNTY, FLORIDA
FOLIO / PARCEL ID:	1828010SVD00000000010U
CODE:	FLORIDA BUILDING CODE RESIDENTIAL 2023 (8TH EDITION)

DESIGN LOADS

LIVE LOAD (LL) = 40 PSF

DEAD LOAD (DL) = 20 PSF

TOTAL LOAD = LL + DL = 40 + 20 = **60 PSF**

TRIBUTARY WIDTH

TRIBUTARY WIDTH = JOIST SPAN ÷ 2

= 13 FT ÷ 2 = **6.5 FT**

LINE LOAD ON BEAM

W = TOTAL LOAD × TRIBUTARY WIDTH

W = 60 PSF × 6.5 FT = **390 PLF**

BEAM SPAN (ASSUMED)

L = 8 FT

MAXIMUM BENDING MOMENT

M = (W × L²) ÷ 8

M = (390 × 8²) ÷ 8 = (390 × 64) ÷ 8

M = **3,120 LB-FT = 37,440 LB-IN**

BEAM SECTION PROPERTIES

(4) 2X8 BUILT-UP BEAM

b = 6.0 IN, d = 7.25 IN

SECTION MODULUS, S = (b × d²) ÷ 6

S = (6.0 × 7.25²) ÷ 6 = **52.6 IN³**

BENDING STRESS CHECK

f_b = M ÷ S

f_b = 37,440 ÷ 52.6 = **712 PSI**

ALLOWABLE F_b (SPF #2) = 875 PSI → **PASS**

SHEAR CHECK

V = (W × L) ÷ 2

V = (390 × 8) ÷ 2 = **1,560 LBS** → PASS

COLUMN LOAD

REACTION PER SUPPORT = V = 1,560 LBS

ADD BEAM SELF-WEIGHT ≈ 200 LBS

TOTAL COLUMN LOAD ≈ **1,800 LBS**

FOOTING SIZE

REQUIRED AREA = LOAD ÷ SOIL PRESSURE

A = 1,800 ÷ 1,500 = **1.2 FT²**

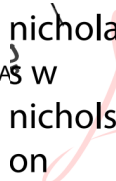
PROVIDE **16" × 16" × 8" CONCRETE FOOTING**

ENGINEER STATEMENT

BUILT-UP BEAMS, COLUMNS, AND FOOTINGS SUPPORTING 13'-0" FLOOR JOISTS ARE DESIGNED BASED ON THE ABOVE STRUCTURAL CALCULATIONS AND SHALL BE REVIEWED AND SEALED BY A FLORIDA-LICENSED PROFESSIONAL ENGINEER.

ENGINEER NAME: NICHOLAS WILLIAM NICHOLSON FL LICENSE NO.: 37862

DATE: JANUARY 29 2026

SIGNATURE & SEAL

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